SQL – Group by & Having

SQL – Group by

SQL – Group by

- **GROUP BY statement groups rows that have the same values in one or more columns.**
- **It is commonly used to create summaries, such as total sales by region or number of users by age group.**

SQL – Group by

- **Used with the SELECT statement.**
- **Groups rows after filtering with WHERE.**
- **Can be combined with aggregate functions like SUM(), COUNT(), AVG(), etc.**
- **Filter grouped results using the HAVING clause.**
- **Comes after WHERE but before HAVING and ORDER BY.**

SQL – Group by

⚠ **Rules**

- All non-aggregated columns in the `SELECT` list **must appear in the** `GROUP BY`.
- You **cannot** use `WHERE` to filter results based on aggregate values — that's what `HAVING` is for.

SQL – Group by

- **Syntax**

Select Column1, aggregate_function(column2)

From table_name

Where condition

GROUP BY column1, column2

SQL – Group by

- **Example**

Student (ID, Name, Department)

Select Department, Count(*) as Student_count

From Student

Group By Department

SQL – Group by

- **Example**

ID	**Name**	**Department**
1	Ahsan Jawed	Computer Science
2	Sara Ahmed	Software Engineering
3	Bilal Khan	Computer Science
4	Fatima Noor	Data Science
5	Ali Raza	Software Engineering
6	Maria Hassan	Data Science

Department	**Student_Count**
Computer Science	2
Software Engineering	2
Data Science	2

SQL – Group by

StudentName	Year	Department
Ahsan Jawed	Senior	Computer Science
Bilal Khan	Junior	Computer Science
Fatima Noor	Senior	Computer Science
Sara Ahmed	Sophomore	Humanities
Ali Raza	Junior	Humanities
Maria Hassan	Sophomore	Humanities

Select Department, Year, Count(*) as Student_count From Student Group By Department, Year

SQL – Group by

Department	Year	Student_Count
Computer Science	Junior	1
Computer Science	Senior	2
Humanities	Junior	1
Humanities	Sophomore	2

SQL – Having

SQL – Having

- **HAVING clause filters results after applying aggregate functions. Unlike WHERE, which filters individual rows, HAVING works with aggregated results.**

SQL – Having

- **Filters results based on aggregate functions.**
- **Supports Boolean conditions (AND, OR).**
- **Applied after aggregate functions in the query.**
- **Useful for summary-level or aggregated filtering.**

SQL – Having

- **Syntax**

```
SELECT AGGREGATE_FUNCTION(column_name)
FROM table_name
HAVING condition;
```

SQL – Having

SELECT column_name1, aggregate_function (column_name2)

FROM table_name

GROUP BY column_name1

HAVING condition;

SQL – Having - Example

Name	Department	Marks
Ahsan Jawed	Computer Science	85
Sara Khan	Computer Science	92
Bilal Ahmed	Information Tech	78
Fatima Ali	Information Tech	88
Hamza Raza	Business Admin	67
Maryam Noor	Business Admin	74
Zain Sheikh	Computer Science	59
Hira Imran	Information Tech	95

```
SELECT Department,
AVG(Marks) AS AvgMarks
FROM Student
GROUP BY Department;
```

Department	AvgMarks
Computer Science	78.67
Information Tech	87.00
Business Admin	70.50

SQL – Having - Example

Name	Department	Marks
Ahsan Jawed	Computer Science	85
Sara Khan	Computer Science	92
Bilal Ahmed	Information Tech	78
Fatima Ali	Information Tech	88
Hamza Raza	Business Admin	67
Maryam Noor	Business Admin	74
Zain Sheikh	Computer Science	59
Hira Imran	Information Tech	95

```
SELECT Department,
AVG(Marks) AS AvgMarks
FROM Student
GROUP BY Department
HAVING AVG(Marks) > 71.0;
```

Department	AvgMarks
Computer Science	78.67
Information Tech	87.00

THANK YOU